

Modules and Packages

A **module** allows you to logically organize your Python code. Grouping related code into a module makes the code easier to understand and use. A module is a Python object with arbitrarily named attributes that you can bind and reference.

Simply, a module is a file consisting of Python code. A module can define functions, classes and variables. A module can also include runnable code.

The import has the following syntax –

```
import module1[, module2[,... moduleN]
```

The from...import * Statement

It is also possible to import all names from a module into the current namespace by using the following import statement –

```
from modname import *
```

Packages in Python

A package is a hierarchical file directory structure that defines a single Python application environment that consists of modules and subpackages and sub-subpackages, and so on.

1. Write a program which returns the year and name of weekday

```
import datetime
x = datetime.datetime.now()
print(x.year)
print(x.strftime("%A"))
```

2. Import datetime module and print current date and time.

```
import datetime
x = datetime.datetime.now()
print(x)
```

3. Write a module for calculator. Perform all the basic mathematical operations using the created module.

```
mod.py
def add(x,y):
    return x+y

def sub(x,y):
    return x-y

def mul(x,y):
    return x*y

def div(x,y):
    return x/y,x%y
```

```
import mod
```

```

x=int(input("Enter a Number:"))
y=int(input("Enter a Number:"))
print(mod.add(x,y))
print(mod.sub(x,y))
print(mod.mul(x,y))
print(mod.div(x,y))

```

4. Import Calendar module and print the calendar of any year.

```

import calendar
print ("The calender of year 2012 is : ")
print (calendar.calendar(2012,2,1,6))

```

5. Write a program which will print the month name and first Monday in a given year.

```

import calendar
yy=int(input("Enter a Year: "))
for month in range (1,13):
    #it retrieves a list of weeks that represent the month
    mycal=calendar.monthcalendar(yy,month)
    week1=mycal[0]
    week2=mycal[1]
    if week1[calendar.Monday]!=0:
        auditday=week1[calendar.Monday]
    else:
        auditday=week2[calendar.Monday]
    print("%10s %2d"%(calendar.month_name[month],auditday))

```

Package

Create a package which contains five modules fish, birds, amphibians, mammals, and reptiles. Each module contains two functions example and characteristics.

```

amphibians.py
def examples():
    print("Here are some examples of amphibians:")
    eggs=['Frog','Salamander','Toads','Newts','caecilians']
    for f in eggs:
        print(f)
def chars():
    ch=["Cold blooded","Lays eggs","Moist scaleless skin"]
    print("Characteristics of amphibians:")
    for c in ch:
        print(c)

```

birds.py

```
def examples():
    print("Here are some examples of birds:")
    eggs=['Parrot','Pigeon','Crow','Owl','Sparrow']
    for f in eggs:
        print(f)
def chars():
    ch=["Have wings","Lays eggs","Warm blooded","Have beaks and no teeth"]
    print("Characteristics of birds:")
    for c in ch:
        print(c)
```

fish.py

```
def examples():
    print("Here are some examples of fish:")
    eggs=['Goldfish','Tuna','Guppy','Piranha','Swordfish']
    for f in eggs:
        print(f)
def chars():
    ch=["Lives Under Water","Breathes through gills","Swims using fins and Tail"]
    print("Characteristics of fish:")
    for c in ch:
        print(c)
```

mammals.py

```
def examples():
    print("Here are some examples of mammals:")
    eggs=["Human","Cat","Tiger","Dog","Elephant"]
    for f in eggs:
        print(f)
def chars():
    ch=["Warm blooded","Gives birth to babies","Have hair and fur","Four chambered heart"]
    print("Characteristics of mammals:")
    for c in ch:
        print(c)
```

reptiles.py

```
def examples():
    print("Here are some examples of reptiles:")
    eggs=['Turtle','Lizard','Crocodiles','Chameleon','Snakes']
    for f in eggs:
        print(f)
def chars():
    ch=["Have scales","Vertebrates","Breathe through lungs","Cold blooded"]
    print("Characteristics of reptiles:")
    for c in ch:
        print(c)
```

```

Creatures.py
import fish,birds,amphibians,mammals,reptiles
def __init__(self):
    print("Find your information about creatures:examples and characteristics")

main.py
#Creatures package includes various modules
import Creatures
print("Find your information about creatures:examples and characteristics")
while(1):
    print("Press\n1 for fish\n2 for birds\n3 for amphibians\n4 for mammals\n5 for reptiles\n6 to exit:")
    c=int(input())
    if c==1:
        Creatures.fish.examples()
        Creatures.fish.chars()
    elif c==2:
        Creatures.birds.examples()
        Creatures.birds.chars()
    elif c==3:
        Creatures.amphibians.examples()
        Creatures.amphibians.chars()
    elif c==4:
        Creatures.mammals.examples()
        Creatures.mammals.chars()
    elif c==5:
        Creatures.reptiles.examples()
        Creatures.reptiles.chars()
    elif c==6:
        print("Thank You!!!")
        break
    else:
        print("Wrong input")

#for each module class concept has been introduced purposefully.
#to be noted same process can be implemented without introducing separate class

```

Assignment:

1. Create a module which contains the functions as stated below.
 - a) $F(x,y)=F(x-y,y)+1$, if $y \leq x$
 - b) $F(n,r)=F(n-1,r)+F(n-1,r-1)$
 - c) $F(n)=F(n/2)+1$ if $n>1$
 - d) $F(M,N)=1$ if $M=0$, or $M \geq N \geq 1$, and $F(M,N)=F(M-1,N)+F(M-1,N-1)$, otherwise.

e) $B(m,x)=m!/(x!(m-x)!)$ where $m>x$,
 $B(0,0)=B(m,0)=1$ and $B(m,x)=B(m,x-1)*[(m-x+1)/x]$

16. Create a module which contains the functions printing several patterns. Each function will take input the number of rows and/or columns as required.

```
*
* *
* * *
* * * *
* * * * *
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

```
  *
 * *
* * *
* * * *
* * * * *
```

Create a module which contains the functions as stated below.

a) $F(x,y)=F(x-y,y)+1$, if $y \leq x$

b) $F(n,r)=F(n-1,r)+F(n-1,r-1)$

c) $F(n)=F(n/2)+1$ if $n>1$

d) $F(M,N)=1$ if $M=0$, or $M \geq N \geq 1$, and $F(M,N)=F(M-1,N)+F(M-1,N-1)$, otherwise.

e) $B(m,x)=m!/(x!(m-x)!)$ where $m>x$,

$B(0,0)=B(m,0)=1$ and $B(m,x)=B(m,x-1)*[(m-x+1)/x]$

module_fun.py

```
def f1(x,y):
```

```
    if y<=x:
```

```
        return f1(x-y,y)+1
```

```
    return 1
```

```
def f2(n,r):
```

```
    if n>0 and r>0:
```

```
        return (f2(n-1,r)+f2(n-1,r-1))
```

```
    else:
```

```
        return n+r
```

```
def f3(n):
```

```
    if n>1:
```

```
        return f3(n/2)+1
```

```
    return 0
```

```
def f4(m,n):
```

```
    if m==0 or (m>=n and n>=1):
```

```
        return 1
```

```
    else:
```

```
        return f4(m-1,n)+f4(m-1,n-1)
```

```
def fact(n):
```

```
    p=1
```

```
    for i in range(1,n+1):
```

```
        p*=i
```

```
    return p
```

```
def f5(m,x):
```

```
    if x==0:
```

```
        return 1
```

```
    if m>x:
```

```
        return (fact(m)/(fact(x)*fact(m-x)))
```

```
    else:
```

```
        return (f5(m,x-1)*((m-x+1)/x))
```

main.py

```
import module_fun as mf
```

```
while(1):
```

```
    print("Press i to access function number i(eg:-1 for fun1)\nPress 6 to exit")
```

```
c=int(input())
if c==1:
    print("Enter x and y:")
    x,y=input().split()
    x=int(x)
    y=int(y)
    print("Result=",mf.f1(x,y))
elif c==2:
    print("Enter n and r:")
    n,r=input().split()
    n=int(n)
    r=int(r)
    print("Result=",mf.f2(n,r))
elif c==3:
    n=int(input("Enter n:"))
    print("Result=",mf.f3(n))
elif c==4:
    print("Enter m and n")
    m,n=input().split()
    m=int(m)
    n=int(n)
    print("Result=",mf.f4(m,n))
elif c==5:
    print("Enter m and x")
    m,x=input().split()
    m=int(m)
    x=int(n)
    print("Result=",mf.f5(m,x))
elif c==6:
    print("Thanks")
    break
else:
    print("Wrong input")
```